

AWS DataSync – Copy Data from Amazon S3 to Amazon EFS

To Begin with the Lab

Summary of the Lab

In this lab, **AWS DataSync** was used to copy data from an **Amazon S3** bucket to **Amazon EFS**. The first part of the setup created a new S3 bucket, uploaded test files into it, and then created an EFS file system in the same Region. A Linux **EC2** instance was launched to mount the EFS file system and confirm that it was empty before the transfer. After that, a DataSync task was created with **Amazon S3** as the source and **Amazon EFS** as the destination, using an automatically generated IAM role. The transfer was then started and verified through the DataSync execution history and by checking that the copied files appeared inside the mounted EFS directory.

Understanding AWS DataSync

AWS DataSync is an online data transfer service that simplifies moving data between storage systems such as **Amazon S3**, **Amazon EFS**, **Amazon FSx**, and on-premises file servers. It exists so data migration, backup, archiving, and in-cloud processing can happen faster and with less manual work. In this lab, DataSync was used to move objects from an S3 bucket into an EFS file system without copying files manually. The benefit is that AWS handles the transfer securely and efficiently, while you focus on where the data should go instead of building your own copy process.

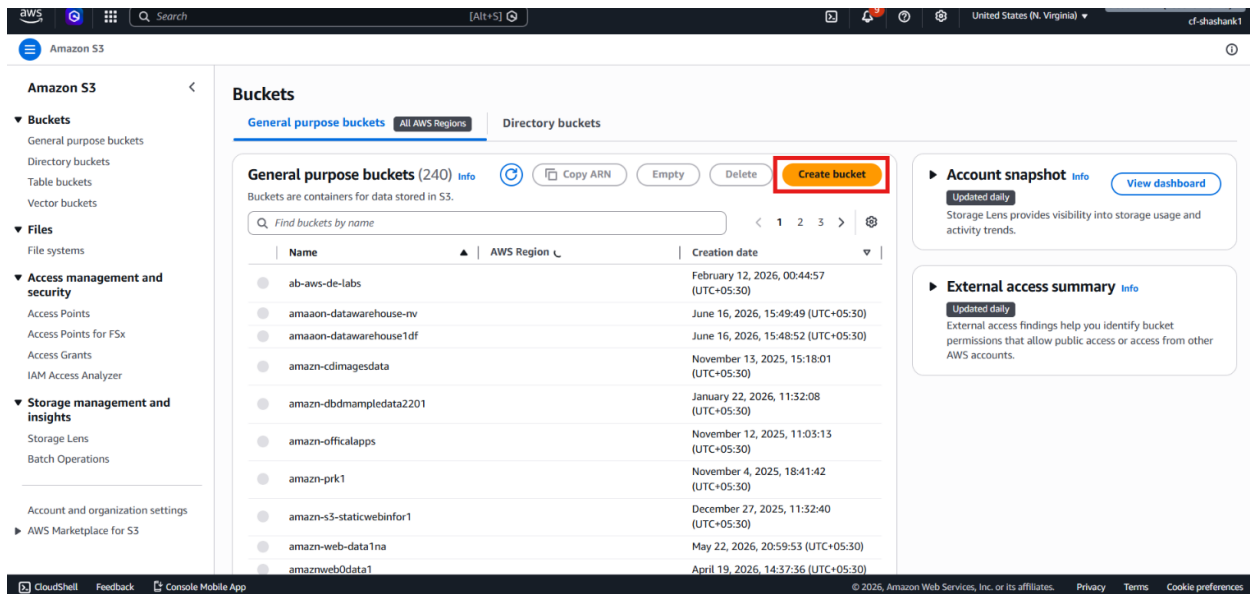
Example:

A project stores 9 image files in an S3 bucket named data sync test 02. The team wants the same files to appear in an EFS file system so a Linux application can read them locally. DataSync moves the files from S3 to EFS, and the files then appear in the mounted EFS directory on the EC2 test instance.

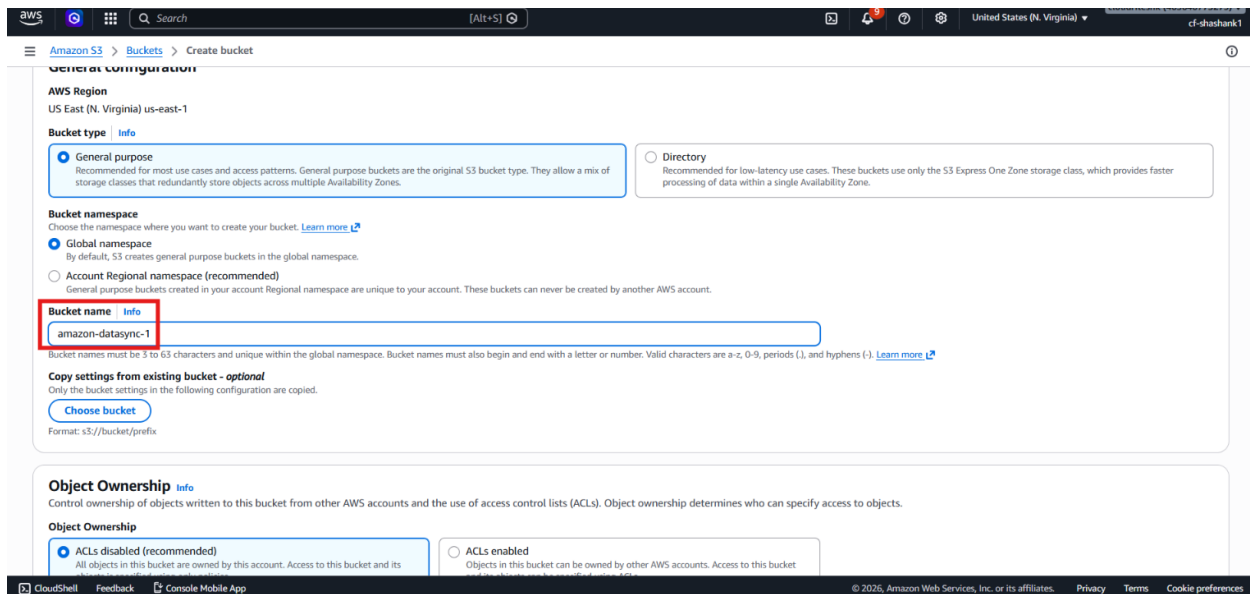
Lab Steps

Step 1 – Create the source S3 bucket

- Sign in to the **AWS Management Console**.
- Open **Amazon S3**.
- Click **Create bucket**.



- Enter the bucket name: **amazon-datasync-1**.
- Select the **Asia Pacific Region**.
- Keep the default public access settings.
- Click **Create bucket**.



- Upload a few test files into the bucket.

Files and folders (3 total, 2.0 MB)

All files and folders in this table will be uploaded.

Find by name

☐	Name	Folder	Type	Size
☐	wallpaper.png	-	image/png	2.0 MB
☐	index.html	-	text/html	5.0 KB
☐	index-india.html	-	text/html	810.0 B

Destination info

Destination

s3://amazon-datasync-1

► Destination details

Bucket settings that impact new objects stored in the specified destination.

► Permissions

Grant public access and access to other AWS accounts.

► Properties

Specify storage class, encryption settings, tags, and more.

Cancel Upload

Step 2 – Create the destination Amazon EFS file system

- Open Amazon EFS.
- Click Create file system.

Amazon Elastic File System

Scalable, elastic, cloud-native NFS file system

Amazon Elastic File System (Amazon EFS) provides a simple, scalable, elastic file system for general purpose workloads for use with AWS Cloud services and on-premises resources.

Create file system

Create an EFS file system with recommended settings.

Create file system

Pricing

With EFS, there are no minimum fees. You pay only for the storage that you use, the data that you read and write, and any additional throughput that you provision. Estimate your cost using the [AWS Pricing Calculator](#).

Learn more about pricing

Get started

What is Amazon Elastic File System? [Get started](#)

- Enter the file system name **Test**.
- Now click on **Customize**.

Create file system

Create a file system with the recommended settings shown below by choosing Create file system. To view all settings or to customize your file system, choose Customize. [Learn more](#)

Name - optional
Name your file system

Name can include letters, numbers, and +, -, / symbols, up to 256 characters.

Virtual Private Cloud (VPC)
Choose the VPC where you want EC2 instances to connect to your file system.
vpc-099792461c7d0a616
default

Recommended settings
Your file system is created with the following recommended settings unless you choose to customize the file system. You will be charged for storage and throughput. We recommend reviewing pricing for these features using the [AWS Pricing Calculator](#).

Setting	Value	Editable after creation
Throughput mode Learn more	Elastic	Yes
Transition into Infrequent Access (IA)	30 day(s) since last access	Yes
Transition into Archive	90 day(s) since last access	Yes
Transition into Standard	None	Yes
Automatic backups	Enabled	Yes
Encryption	Enabled	No

Cancel **Customize** Create file system

- Keep the default **VPC** and customize only the required options.
- Leave encryption at rest disabled for this lab.
- Keep automated backup off.

Amazon EFS > File systems > Create

Step 3 - optional
File system policy
Step 4
Review and create

Name - optional
Name your file system

Name can include letters, numbers, and +, -, / symbols, up to 256 characters.

File system type
Choose to either store data across multiple Availability Zones or within a single Availability Zone. [Learn more](#)

☒ **Regional**
Provides the highest levels of availability and durability by storing file system data across multiple Availability Zones within an AWS Region.

☐ **One Zone**
Provides continuous availability to data within a single Availability Zone within an AWS Region.

Automatic backups
Automatically backup your file system's data with AWS Backup using recommended settings. Additional pricing applies. [Learn more](#)

☐ **Enable automatic backups**

Lifecycle management
Automatically save money as access patterns change by moving files into the Infrequent Access (IA) or Archive storage class. [Learn more](#)

Transition into Infrequent Access (IA)
Transition files to IA based on the time since they were last accessed in Standard storage.
30 day(s) since last access

Transition into Archive
Transition files to Archive based on the time since they were last accessed in Standard storage.
90 day(s) since last access

Transition into Standard
Transition files back to Standard storage based on when they are first accessed in IA or Archive storage.
None

Encryption
Choose to enable encryption of your file system's data at rest. Uses the AWS KMS service key (aws/elasticfilesystem) by default. [Learn more](#)

☐ **Enable encryption of data at rest**

- Mount the file system in **ap-south-1a** and **ap-south-1b** but I am using **N.Virginia** so I will select respective availability zones.
- Use the default security group.

Step 2
 Network access
 Step 3 - optional
 File system policy
 Step 4
 Review and create

Network
 Virtual Private Cloud (VPC) [Learn more](#)
 Choose the VPC where you want EC2 instances to connect to your file system.
 vpc-099792461c7d0a616
 default

Mount targets
 A mount target provides an NFSv4 endpoint at which you can mount an Amazon EFS file system. We recommend creating one mount target per Availability Zone. [Learn more](#)

Availability zone	Subnet ID	IP address type	IPv4 address	IPv6 address	Security groups
us-east-1a	subnet-0b5...	IPv4 only	Optional	-	sg-09ba64237d6 default
us-east-1b	subnet-0d1...	IPv4 only	Optional	-	sg-09ba64237d6 default

[Add mount target](#)

[Cancel](#) [Previous](#) [Next](#)

- Click through the remaining steps and create the file system.

Step 3 – Launch a Linux EC2 instance to test EFS

- Open Amazon EC2.
- Launch a new Amazon Linux instance.

Name
 EFS_Test [Add additional tags](#)

Application and OS Images (Amazon Machine Image) [Info](#)
 An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).
 Search your full catalog including 1000s of application and OS images

Recents **My AMIs** **Quick Start**

Amazon Linux [Browse more AMIs](#)
 Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)
 Amazon Linux 2023 kernel-6.18 AMI
 ami-01edba92f9036f76e (64-bit (x86), uefi-preferred) / ami-08bee9c1b63b637da (64-bit (Arm), uefi)
 Virtualization: hvm ENA enabled: true Root device type: ebs

Description
 Amazon Linux 2023 (kernel-6.18) is a modern, general purpose Linux-based OS that will be supported until June 2029. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.
 Amazon Linux 2023 AMI 2023.12.20260710.0 x86_64 HVM kernel-6.18

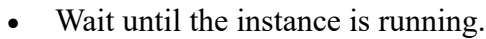
Architecture **Boot mode** **AMI ID** **Publish Date** **Username**
 x86_64 uefi ami-01edba92f9036f76e 2026-07-10 ec2-user [Verified provider](#)

Summary
 Number of instances [Info](#)
 1
 Software image (AMI)
 Amazon Linux 2023 AMI 2023.12....[read more](#)
 ami-01edba92f9036f76e
 Virtual server type (instance type)
 t3.micro
 Firewall (security group)
 New security group
 Storage (volumes)
 1 volume(s) - 8 GiB

[Free tier](#) In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs. 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet. Data transfer charges are not included as part of the free tier allowance. Charges may apply depending on your account's free tier status.

[Cancel](#) [Launch instance](#) [Preview code](#)

- Place the instance in **ap-south-1a** so it matches the EFS mount targets.
- Keep the storage small and use the default SSH security group settings.
- Give the instance a test name and launch it.



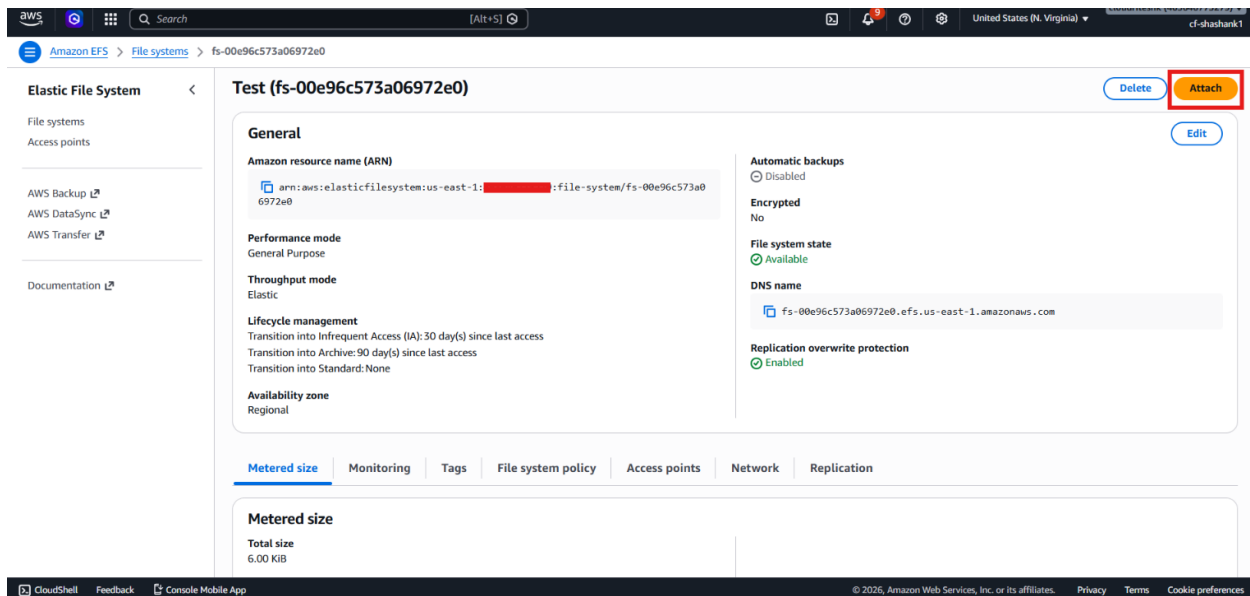
- Connect to the EC2 instance.

- Login as root user.

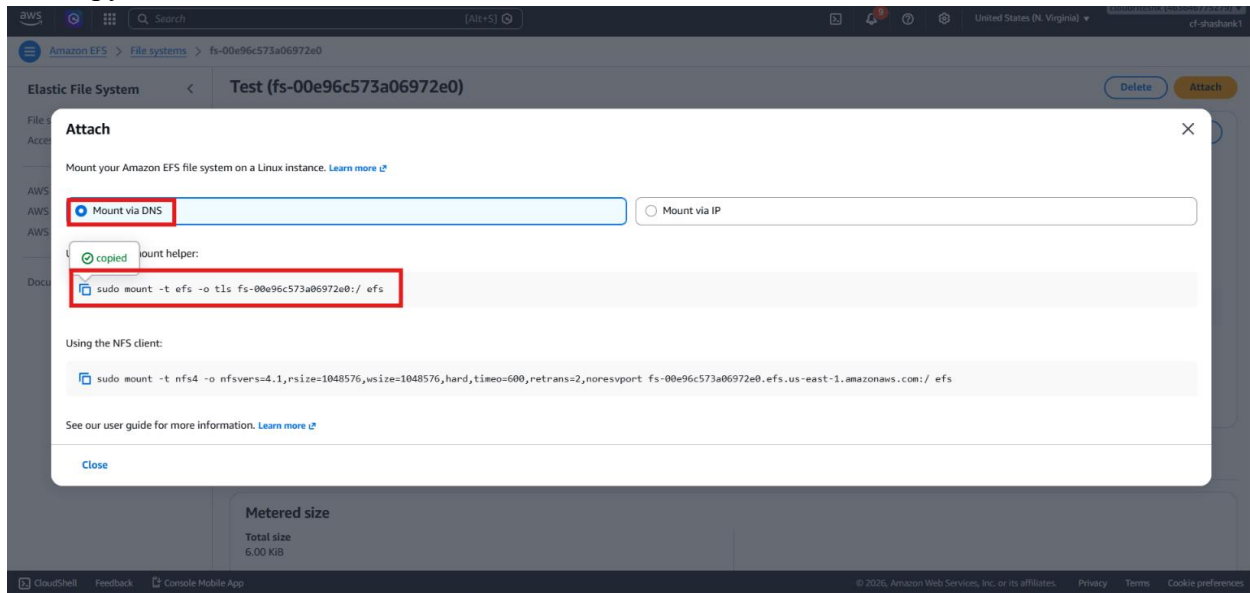
- Create a mount directory named efs.

[illegible]

- Open the EFS file system and click **Attach**.

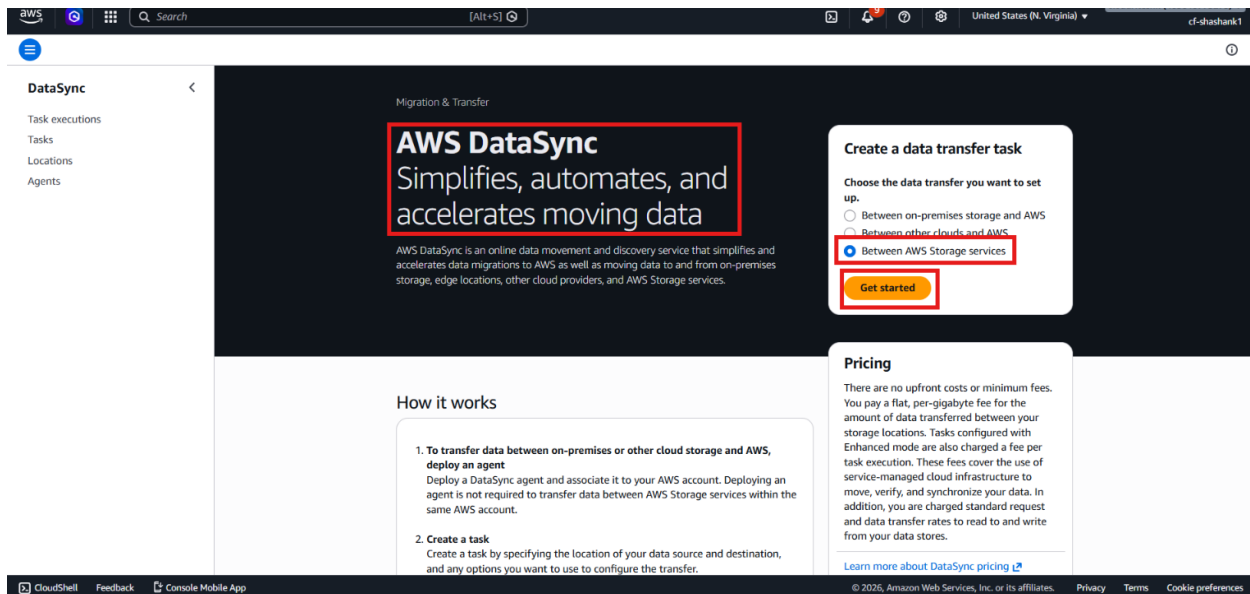


- Copy the mount command.

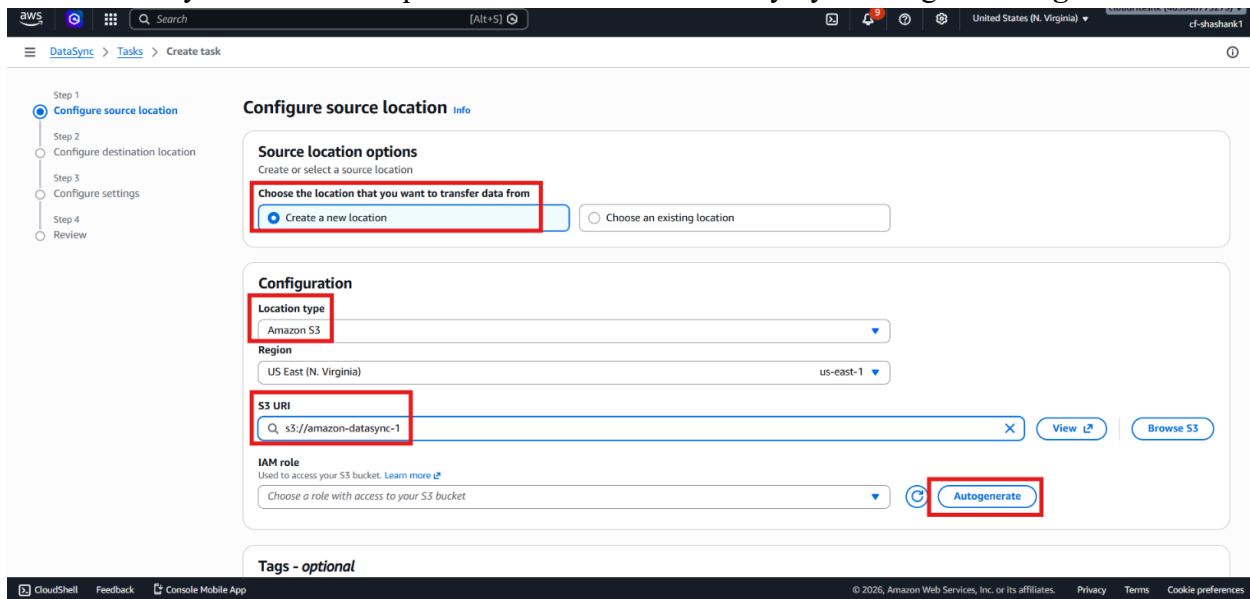


- Run it in the EC2 Instance.
- Install **Amazon EFS Utils** if you get **unknown filesystem type** error.

```
yum install amazon-efs-utils
```

- Select **Amazon S3** as the source location.
- Choose the bucket: **amazon-datasync-1**.
- Let DataSync create the required IAM role automatically by clicking on **Autogenerate**.



- Select **Amazon EFS** as the destination location.
- Choose the file system: **Test**.
- Set the destination path to **/**.
- Choose the **Subnet** and **Security group**.

Configuration

Location type
Amazon EFS

Region
US East (N. Virginia)

File system
Test (fs-00e96c573a06972e0)

Mount path
A subdirectory of the selected file system.
/

Subnet
A subnet that the file system will be accessed from. [Learn more](#)
subnet-0b5783ef03796c671

Security groups
Security group(s) that can access the specified file system. [Learn more](#)
Choose security groups
default (sg-09ba64237d6c846f3)

In-transit encryption
The protocol to use for encryption in-transit.
☒ None
☐ TLS 1.2

- Name the task: Task1.
- Now Choose **Basic** Task mode.
- Choose **Everything** for Contents to scan.
- Set Transfer Mode to **Transfer all data**.

Task mode Info
Your task options and performance might vary depending on the task mode you choose. Task mode can't be changed once you create this task. [See pricing details.](#)

☐ Enhanced
Transfer virtually unlimited numbers of objects with advanced metrics, more detailed logs, and higher performance than Basic mode. Currently available for transfers between S3 buckets, between S3 and other clouds when not using an agent, or between NFS or SMB servers and S3 using an Enhanced mode agent.

☒ **Basic**
Subject to quotas on the number of files and objects in a dataset. Lower performance than Enhanced mode for most workloads. Requires an agent when copying to or from on-premises or other clouds.

Name - optional
Task1

Source data options Info
Contents to scan
Specify the data in your source location that you want to transfer to your destination.
Everything

Excludes - optional Info
Specify the files, objects, and folders that you don't want to transfer.
/folder-to-exclude or *temp-file [Add pattern](#)

Transfer options
Transfer mode
Determines whether DataSync transfers only the data and metadata that differ between the source and the destination.
Transfer all data

- Scroll down and Choose **Autogenerate** for the CloudWatch Resource Policy.

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. No tags associated with the resource.

[Add new tag](#)

You can add up to 50 tags.

Task report - optional

Get a detailed report for your DataSync transfer task, including specific file names and how much data you moved. Amazon S3 pricing applies. [Learn more](#)

Report type

None

Logging

Log level

Defines the level of detail that is logged to CloudWatch Logs. CloudWatch Logs pricing applies for logs generated by DataSync.

Log basic information such as transfer errors

CloudWatch log group

Log task execution activity such as errors or other failures.

/aws/datsync

☒ Create a CloudWatch resource policy

[Autogenerate](#)

[Cancel](#) [Previous](#) [Next](#)

- Create the task.

Step 6 – Start and verify the transfer

- Start the DataSync task.
- Open the **Task1** and click on **Start with defaults**.

DataSync

Task executions
Tasks
Locations
Agents

Task1

[Start](#) [Stop](#) [Edit](#) [Delete](#)

Start with defaults

Start with overriding options

Creation time

Jul 14, 2026, 3:27:43 PM GMT+5:30

Overview

Task status

Available

Latest execution

-

Task mode

Basic

Task ARN

arn:aws:datsync:us-east-1:██████████:task/task-0a057c216a67597c9

Locations **Options** **Schedule** **Tags** **Monitoring** **Task executions**

Source location

Location ID

loc-02d423103c530474f

Region

US East (N. Virginia) us-east-1

Type

Amazon S3

Path

/

Creation Time

Jul 14, 2026, 3:27:42 PM GMT+5:30

S3 bucket

amazon-datsync-1

S3 storage class

Standard

S3 bucket IAM role

AWSDataSyncS3BucketAccess-amazon-datsync-1-5b6f4

Destination location

Location ID

loc-0ea71f1bed7af130f

Region

US East (N. Virginia) us-east-1

Type

Amazon EFS

Path

/

Creation Time

Jul 14, 2026, 3:27:42 PM GMT+5:30

File system

fs-00e96c573a06972e0

Subnet

subnet-0b5783ef03796c671

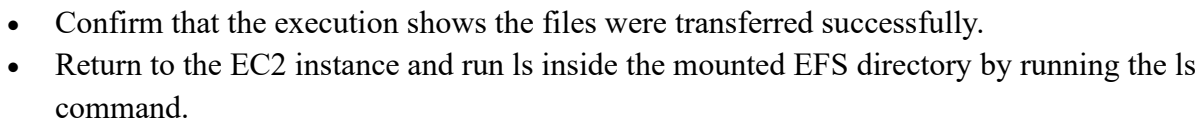
Security groups

- sg-09ba64237d6c846f3

In-transit encryption

None

- Wait for the transfer to complete.
- Open the task execution history.



- Verified that the source **Amazon S3** bucket was created successfully.
- Verified that test files were uploaded to the S3 bucket.
- Verified that the **Amazon EFS** file system was created successfully.
- Verified that the Linux **EC2** instance mounted the EFS file system correctly.
- Verified that the **AWS DataSync** task was created with S3 as the source and EFS as the destination.
- Verified that the transfer completed successfully.
- Verified that the copied files appeared in the mounted EFS directory.